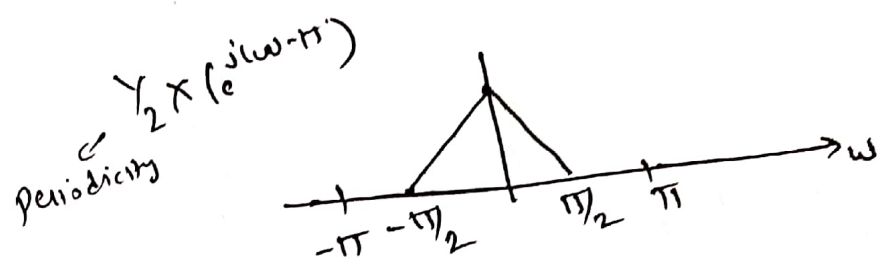
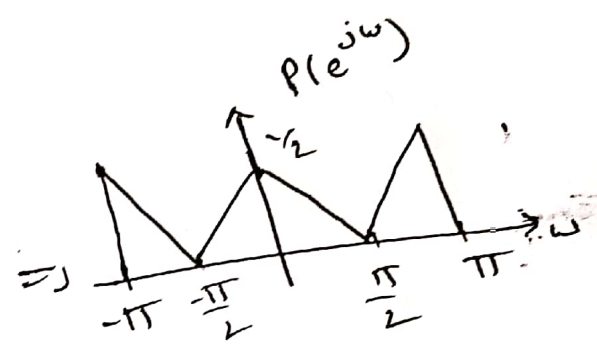
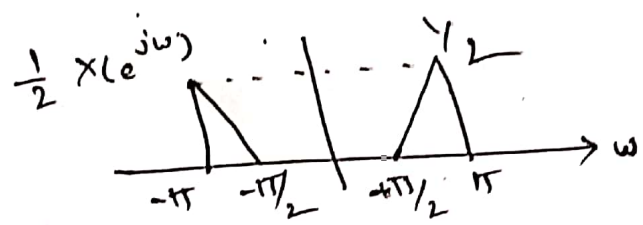


$$P(z) = \frac{1}{2} \sum_{k=0}^1 x(z^{2/2} \omega_2^k) = \frac{1}{2} \sum_{k=0}^1 x(z \cdot e^{-j2\pi k/2})$$

$$= \frac{1}{2} \sum_{k=0}^1 x(z \cdot e^{-j\pi k}) = \frac{1}{2} [x(z) + x(e^{-j\pi} \cdot z)]$$

$$P[e^{j\omega}] = \frac{1}{2} [x(e^{j\omega}) + x(e^{j(\omega-\pi)})] \quad \text{--- (2)}$$



$$\therefore y_0(e^{j\omega}) = P(e^{j\omega}) \cdot H_0(e^{j\omega}) =$$

$$y_1(e^{j\omega}) = P(e^{j\omega}) \cdot H_1(e^{j\omega}) =$$

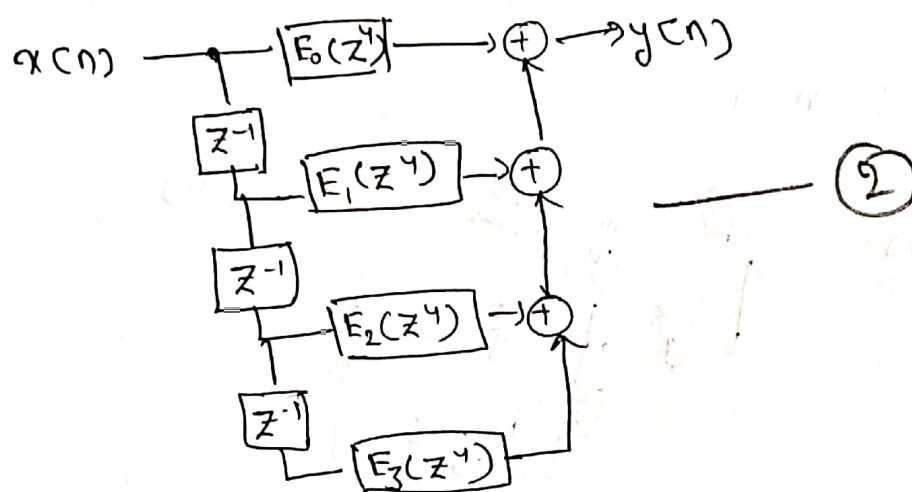
(b) For down-sampled signal, it's still possible to recover the original signal using filters and multi-rate building blocks as long as there is no aliasing [as $H_1(e^{j\omega})$ in the above question] --- (2)

$$H(z) = R + Tz^{-1} + 2Rz^{-2} + 2Tz^{-3} + 2Tz^{-4} + 2Rz^{-5} + Tz^{-6} + Rz^{-7}$$

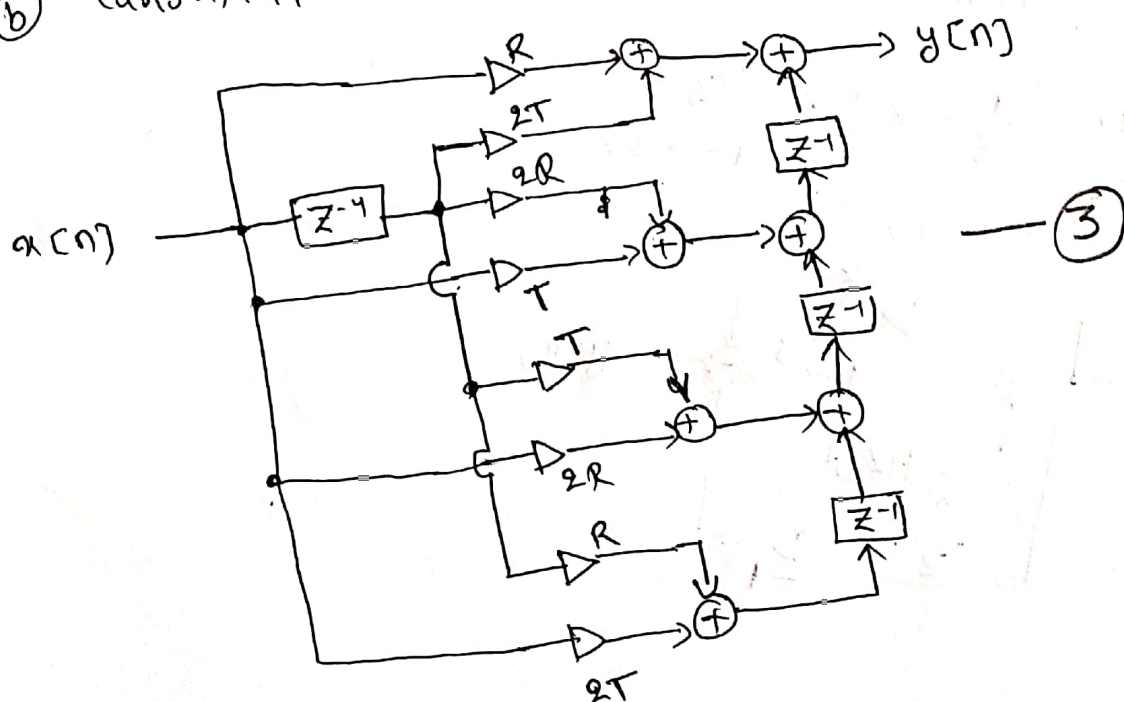
① $H(z) = F_0(z^4) + z^{-1}E_1(z^4) + z^{-2}E_2(z^4) + z^{-3}E_3(z^4)$

$$\left. \begin{aligned} E_0(z) &= R + 2Tz^{-1}, & E_1(z) &= T + 2Rz^{-1} \\ E_2(z) &= 2R + Tz^{-1}, & E_3(z) &= 2T + Rz^{-1} \end{aligned} \right\} \text{--- ②}$$

4-band polyphase realization $H(z)$



⑥ canonical Realization $\Rightarrow$ 7-delays



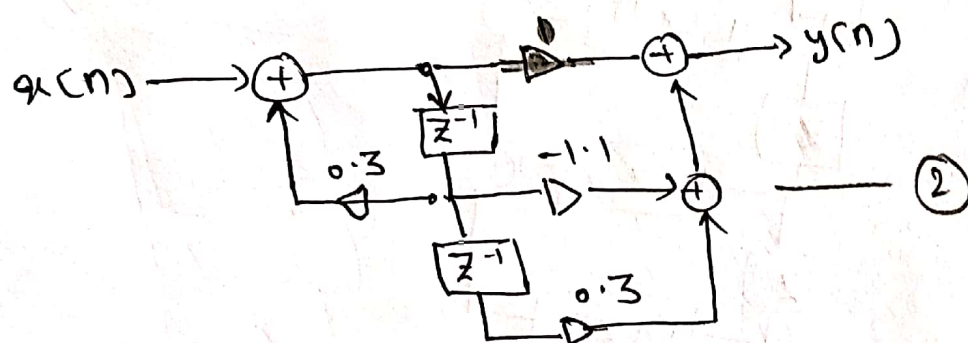
$$Y(z) = \frac{1}{1-0.3z^{-1}} - \frac{0.5z^{-1}}{1-0.3z^{-1}} \quad ; \quad Y(z) = \frac{1-0.5z^{-1}}{1-0.3z^{-1}}$$

$$X(z) = \frac{1}{1-0.6z^{-1}} \quad ; \quad H(z) = \frac{Y(z)}{X(z)} = \frac{(1-0.5z^{-1})(1-0.6z^{-1})}{1-0.3z^{-1}}$$

$$= \frac{1 - 1.1z^{-1} + 0.3z^{-2}}{1-0.3z^{-1}} \quad \text{--- (1)}$$

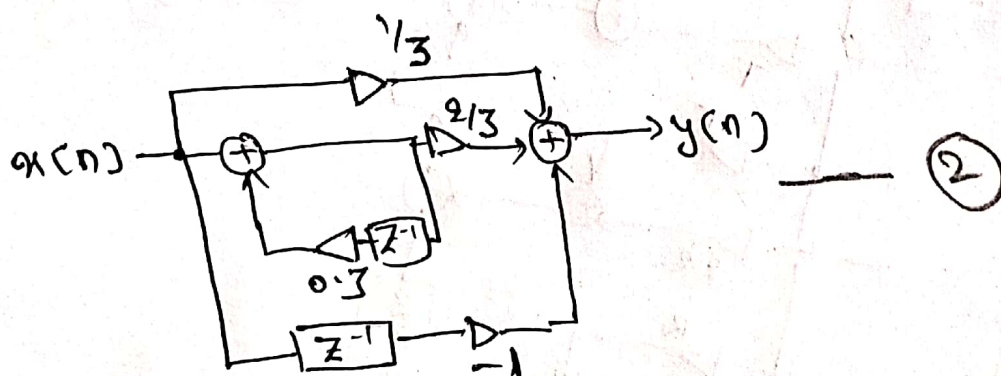
$$y[n] = x[n] - 1.1x[n-1] + 0.3x[n-2] + 0.3y[n-1]$$

Direct form-II --- (1)



Parallel form-I realization

$$H(z) = \frac{1}{3} - z^{-1} + \frac{2/3}{1-0.3z^{-1}} \quad \text{--- (1)}$$



(6)

$$\delta_p = 0.8 ; \delta_s = 0.2 ; \omega_p = 0.2\pi ; \omega_s = 0.6\pi, T=1$$

$$\epsilon = 0.75$$

pre-warping.

$$\Omega_p = 2 \tan\left(\frac{\omega_p}{2}\right) = 2 \tan(0.1\pi) = 0.6498 \text{ rad/sec}$$

$$\Omega_s = 2 \tan\left(\frac{\omega_s}{2}\right) = 2 \tan(0.3\pi) = 2.752 \text{ rad/sec}$$

$$N = \frac{\cosh^{-1} \left[\frac{(\Omega_p/0.2)^2 - 1}{(\Omega_p/0.8)^2 - 1} \right]^{1/2}}{\cosh^{-1} \left(\frac{2.752}{0.6498} \right)} = 1.208 \quad \therefore N=2 \quad \text{--- (1)}$$

LPT $\Omega_p = 0.6498 \text{ rad/sec}$

$$H_c(s) = \frac{\Omega_p^2}{s^2 + b_1 \Omega_p s + c_1 \Omega_p^2}$$

$$y_M = \frac{1}{2} \left\{ \sqrt{1 + \frac{1}{\epsilon^2}} \dots \right\} = 0.5774 \quad c_1 = 0.8333 \quad b_1 = 0.8165 \quad \text{--- (1)}$$

$$H(s) = \frac{0.2815}{s^2 + 0.5306s + 0.3519} \quad \text{--- (2)}$$

Using BLT

$$H(z) = H(s) \Big|_s = \frac{2}{T} \left[\frac{1 - z^{-1}}{1 + z^{-1}} \right]$$

$$T = 1 \text{ sec}$$

$$H(z) = \frac{0.052 (1 + z^{-1})^2}{1 - 1.3480 z^{-1} + 0.608 z^{-2}} \quad \text{--- (2)}$$

$$V_0(z) = \begin{bmatrix} z^{-1} \\ 1+z^{-1} \\ 1 \end{bmatrix} x(z) ; w(z) = \begin{bmatrix} z^{-1/2} \\ 1+z^{-1/2} \\ 1 \end{bmatrix} x(z^{1/2}) +$$

$$\begin{bmatrix} -z^{-1/2} \\ 1-z^{-1/2} \\ 1 \end{bmatrix} x(-z^{1/2}) \quad \text{--- (1)}$$

$$\begin{bmatrix} R_0(z) \\ R_1(z) \\ R_2(z) \end{bmatrix} R(z) = \begin{bmatrix} z^{-1} H_0(z^2) \\ (1+z^{-1}) \{ H_0(z^2) + H_1(z^2) \} \\ H_1(z^2) \end{bmatrix} x(z) + \begin{bmatrix} -z^{-1} H_0(z^2) \\ (1-z^{-1}) \{ H_0(z^2) + H_1(z^2) \} \\ H_1(z^2) \end{bmatrix} x(-z) \quad \text{--- (1)}$$

$$Y(z) = (1-z^{-1}) R_0(z) + z^{-1} R_1(z) + (z^{-2} - z^{-1}) R_2(z)$$

$$Y(z) = 2z^{-1} [H_0(z^2) + z^{-1} H_1(z^2)] x(z)$$

$$W(z) = 2z^{-1} [H_0(z^2) + z^{-1} H_1(z^2)] \quad \text{--- (2)}$$

$$\textcircled{b} \quad W(z) = 2z^{-1} H(z) \quad \text{--- (2)}$$

$$\therefore H(z) \text{ is all APF} \quad \text{--- (1)}$$